THE DEGREE OF A FINITE LOCALLY FREE SHEAF OF CONSTANT RANK IS INVARIANT UNDER PULLBACK ALONG A BIRATIONAL MORPHISM OF SCHEMES OF DIMENSION AT MOST ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites six cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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Allowed standard prerequisites (editorial).

- A proper morphism with finite fibres is finite. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-proper-finite-fibre-finite-in-neighbourhood (tag 02OH); Morphisms, Lemma morphisms-lemma-image-proper-scheme-closed (tag 01W6). Turns the birational hypothesis into finiteness of f, which is what makes the rest of the computation work.
- Euler characteristics: additivity, the projection formula, and the finite support lemma. Stacks Project, Varieties, Lemmas lemma-euler-characteristic-additive (tag 08AA) and lemma-chi-tensor-finite (tag 0AYT); Cohomology, Lemma cohomology-lemma-projection-formula (tag 01E8). Supply the four equalities of the telescoping computation.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02OH: sources/stacks-chow/tags.txt, source lines 3516–3516. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01E8: sources/stacks-chow/tags.txt, source lines 1825–1825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08AA: sources/stacks-chow/tags.txt, source lines 10525–10525. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYT: sources/stacks-chow/tags.txt, source lines 13801–13801. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

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Lemma 0.1. *Let k be a field. Let $f : X' \to X$ be a birational morphism of proper schemes of dimension ≤ 1 over k. Then*

$$\deg(f^*\mathcal{E}) = \deg(\mathcal{E})$$

for every finite locally free sheaf of constant rank. More generally it suffices if f induces a bijection between irreducible components of dimension 1 and isomorphisms of local rings at the corresponding generic points.

Proof. The morphism f is proper (Morphisms, Lemma 01W6) and has fibres of dimension ≤ 0. Hence f is finite (Cohomology of Schemes, Lemma 02OH). Thus

$$Rf_*f^*\mathcal{E} = f_*f^*\mathcal{E} = \mathcal{E} \otimes_{\mathcal{O}_X} f_*\mathcal{O}_{X'}$$

Since f induces an isomorphism on local rings at generic points of all irreducible components of dimension 1 we see that the kernel and cokernel

$$0 \to \mathcal{K} \to \mathcal{O}_X \to f_*\mathcal{O}_{X'} \to \mathcal{Q} \to 0$$

have supports of dimension ≤ 0. Note that tensoring this with $\mathcal{E}$ is still an exact sequence as $\mathcal{E}$ is locally free. We obtain

$$\begin{aligned}
\chi(X, \mathcal{E}) - \chi(X', f^*\mathcal{E}) &= \chi(X, \mathcal{E}) - \chi(X, f_*f^*\mathcal{E}) \\
&= \chi(X, \mathcal{E}) - \chi(X, \mathcal{E} \otimes f_*\mathcal{O}_{X'}) \\
&= \chi(X, \mathcal{K} \otimes \mathcal{E}) - \chi(X, \mathcal{Q} \otimes \mathcal{E}) \\
&= n\chi(X, \mathcal{K}) - n\chi(X, \mathcal{Q}) \\
&= n\chi(X, \mathcal{O}_X) - n\chi(X, f_*\mathcal{O}_{X'}) \\
&= n\chi(X, \mathcal{O}_X) - n\chi(X', \mathcal{O}_{X'})
\end{aligned}$$

which proves what we want. The first equality as f is finite, see Cohomology of Schemes, Lemma 089W. The second equality by projection formula, see Cohomology, Lemma 01E8. The third by additivity of Euler characteristics, see Lemma 08AA. The fourth by Lemma 0AYT. □